MANCHESTER 1824

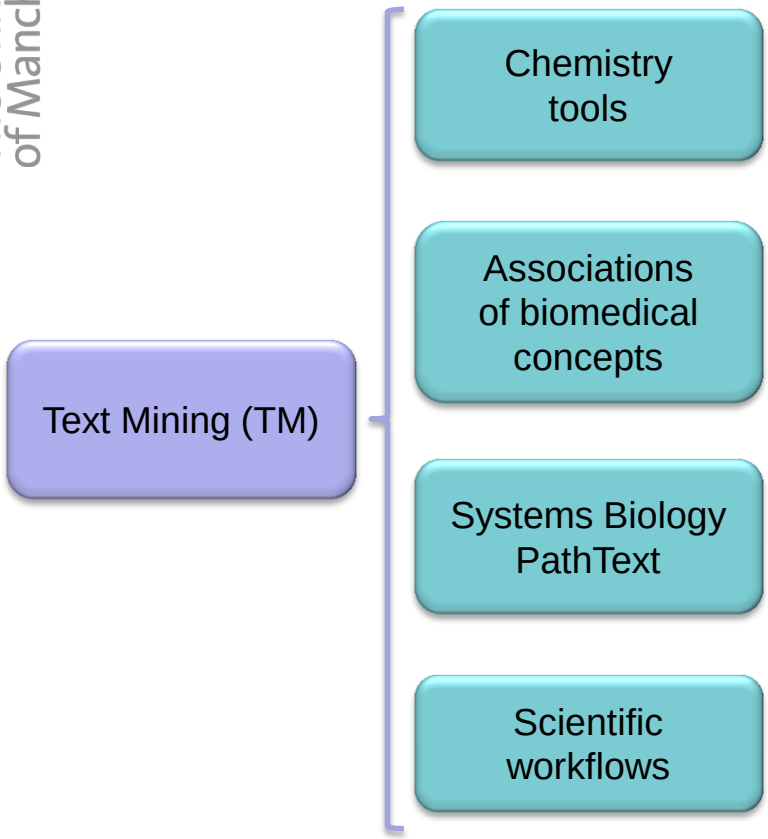
The University of Manchester

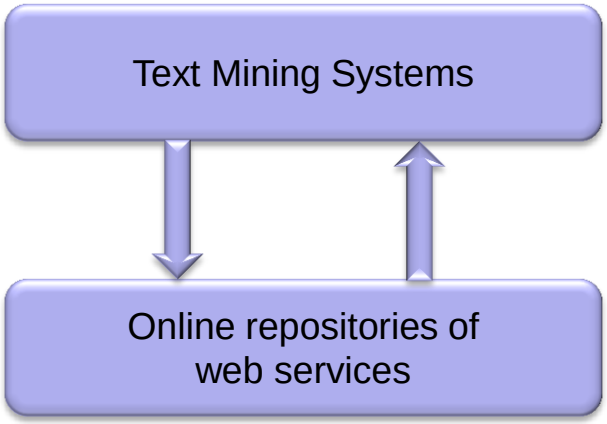
Adding Text Mining Workflows as Web Services to the BioCatalogue

Georgios Kontonatsios, Ioannis Korkontzelos
BalaKrishna Kolluru, Sophia Ananiadou

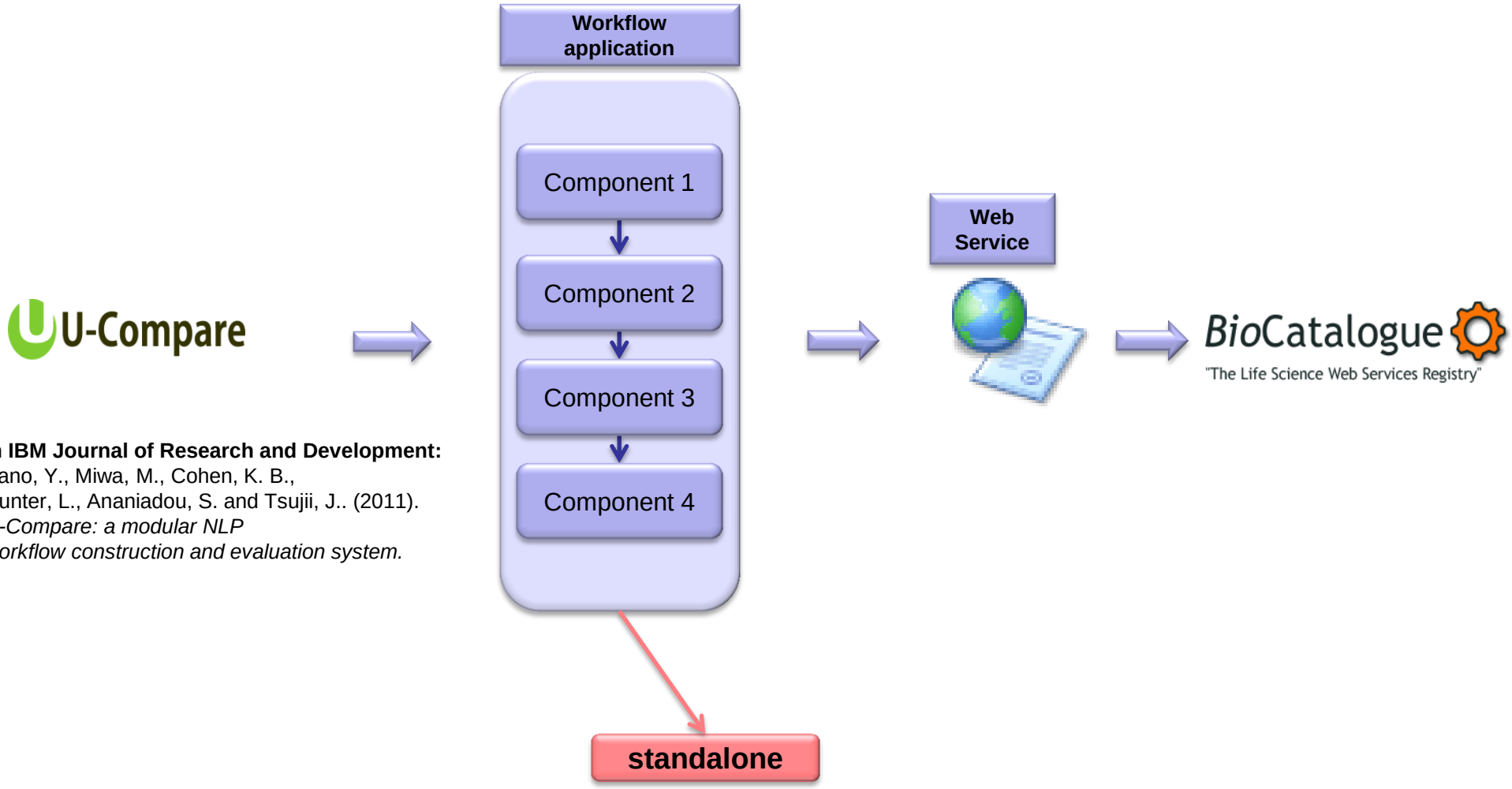
NaCTeM
The National Centre for Text Mining

Building Semantic tools for the Life Sciences using Text Mining





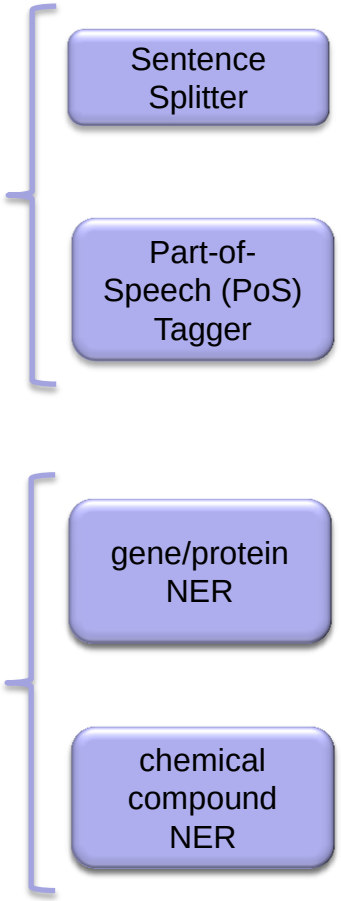
Overview

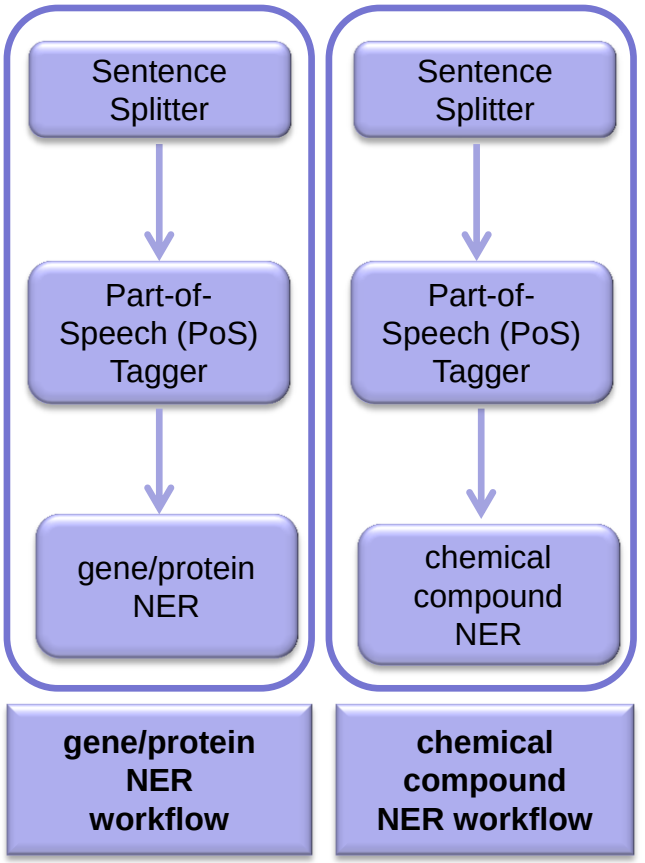


In IBM Journal of Research and Development:
Kano, Y., Miwa, M., Cohen, K. B.,
Hunter, L., Ananiadou, S. and Tsujii, J.. (2011).
U-Compare: a modular NLP
workflow construction and evaluation system.

Background: Why workflows ?

- Individual components may be of little interest
- Others require pre-processing steps



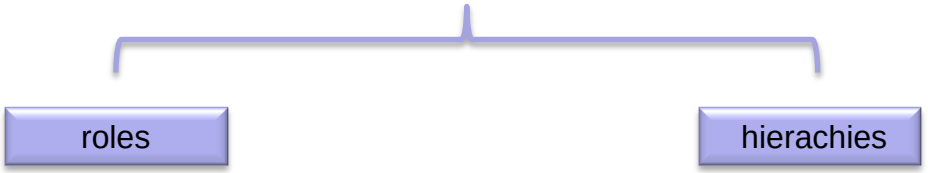


Background: UIMA

- A framework that encourages interoperability in text analysis components



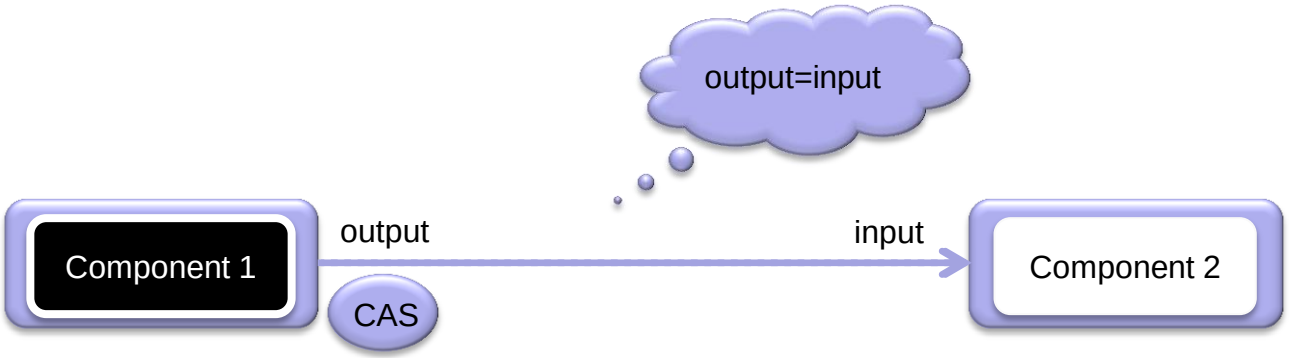
- An architecture that facilitates workflow construction through



Background: U-Compare

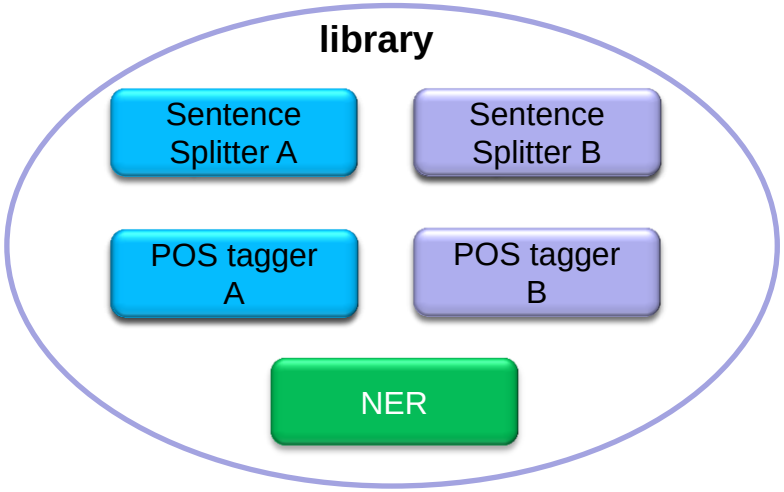
- U-Compare is a TM platform built on top of UIMA

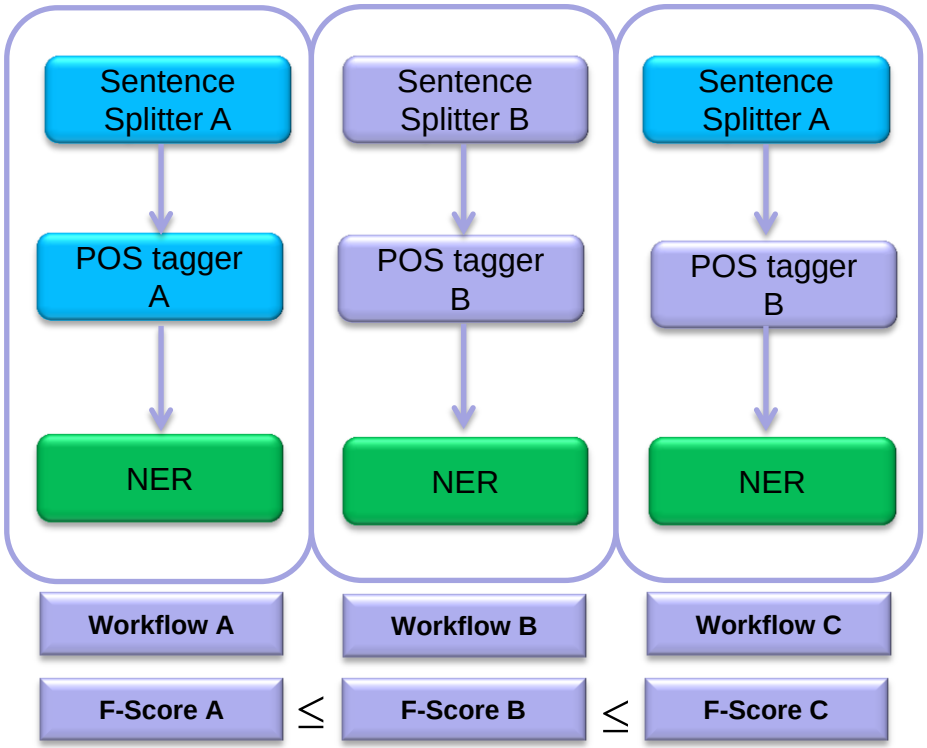
- It enables interoperability by defining a common Type System.



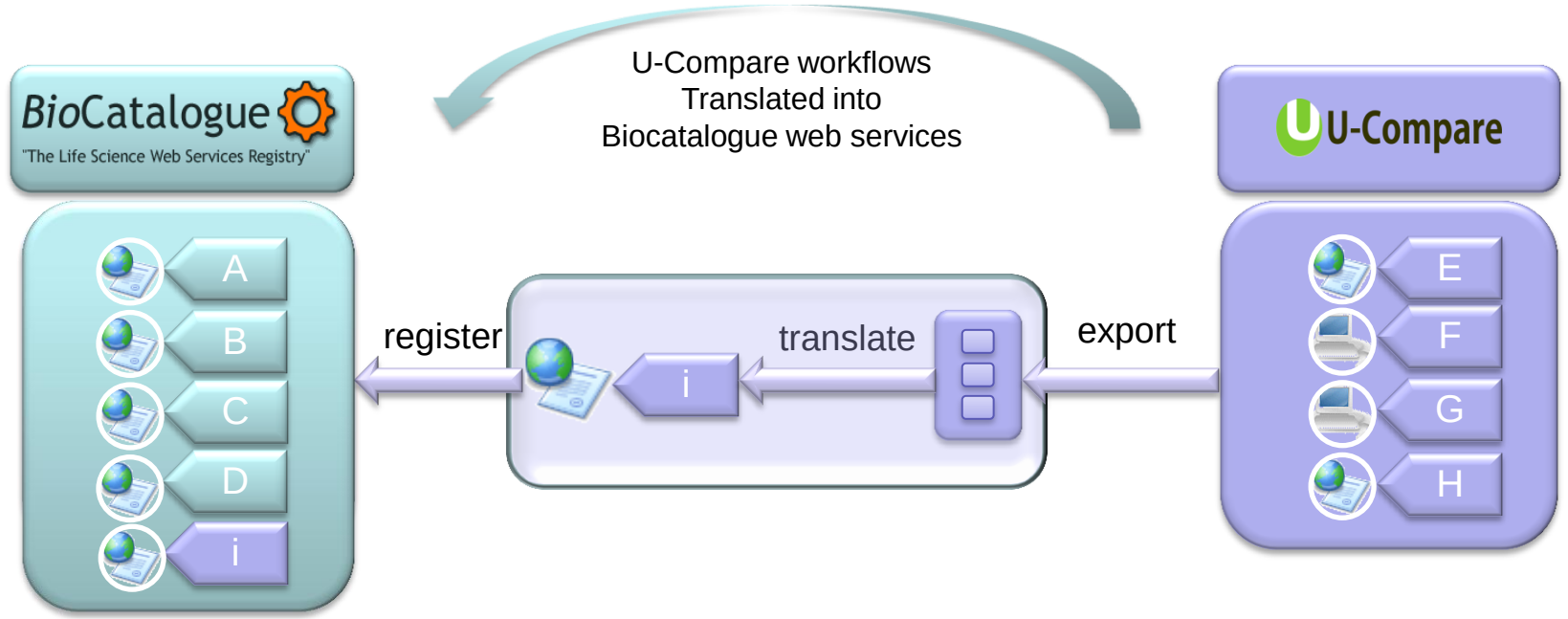
Background: U-Compare

- U-Compare allows the comparison and evaluation of workflows.
  - Critical processes for constructing optimal workflows





Integrating Biocatalogue with U-Compare



- Expose our solutions to the bioinformatics community
- Alternative access mechanism to U-Compare workflows
- Workflows are:
  - independent of programming languages or software libraries.
  - loosely-coupled components that can be reused in other applications.

User requirements and design objectives

developers

- Integrate programmatically with the web services

→ Web API



user

- Well documented web services

→ Documentation tools

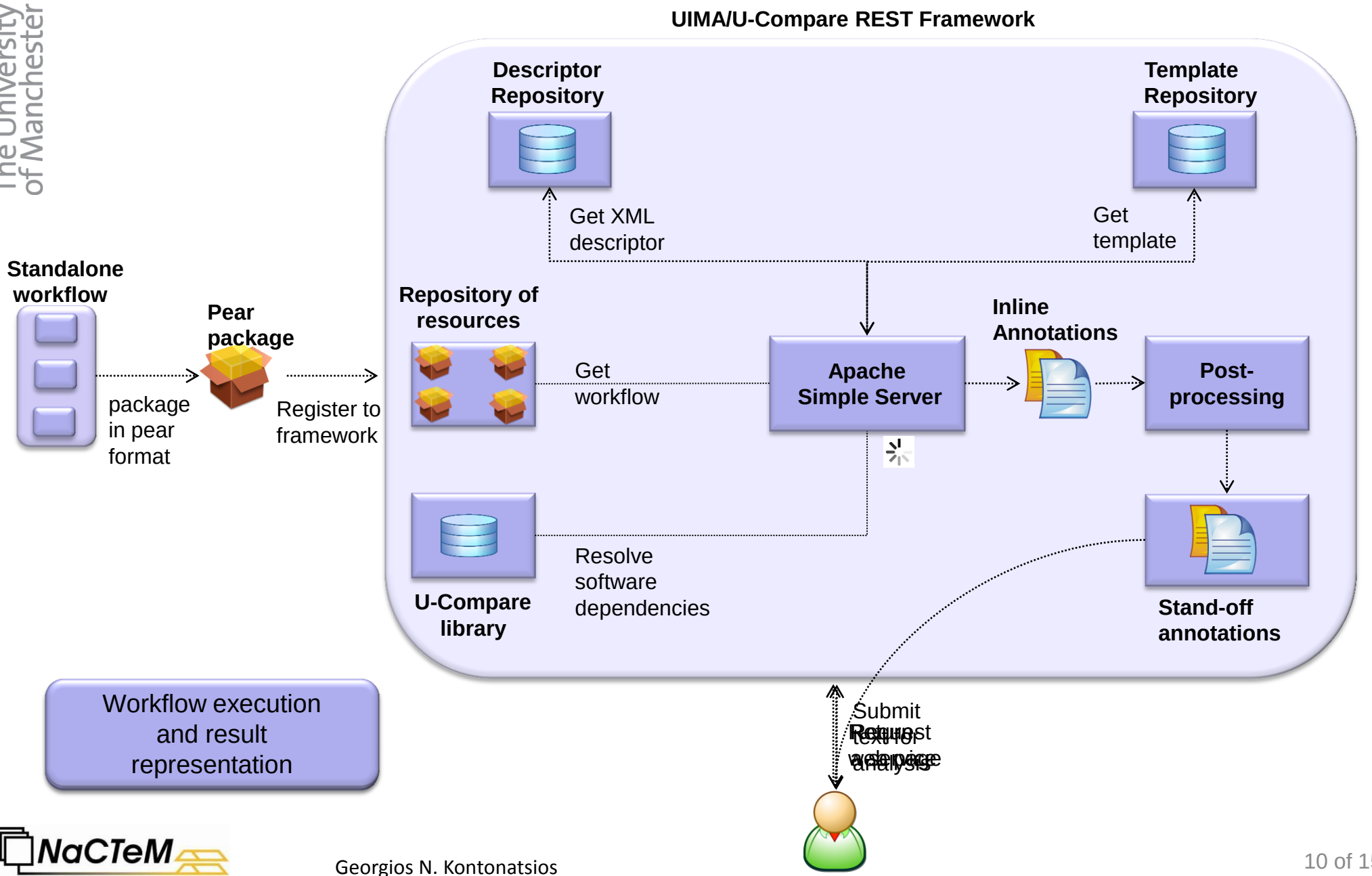
end-users

- easy-to-use tools
- Stay away from markup languages

→ Human readable access mechanism

→ Visualization of results

Architecture of the web application framework



Architecture of the web application framework

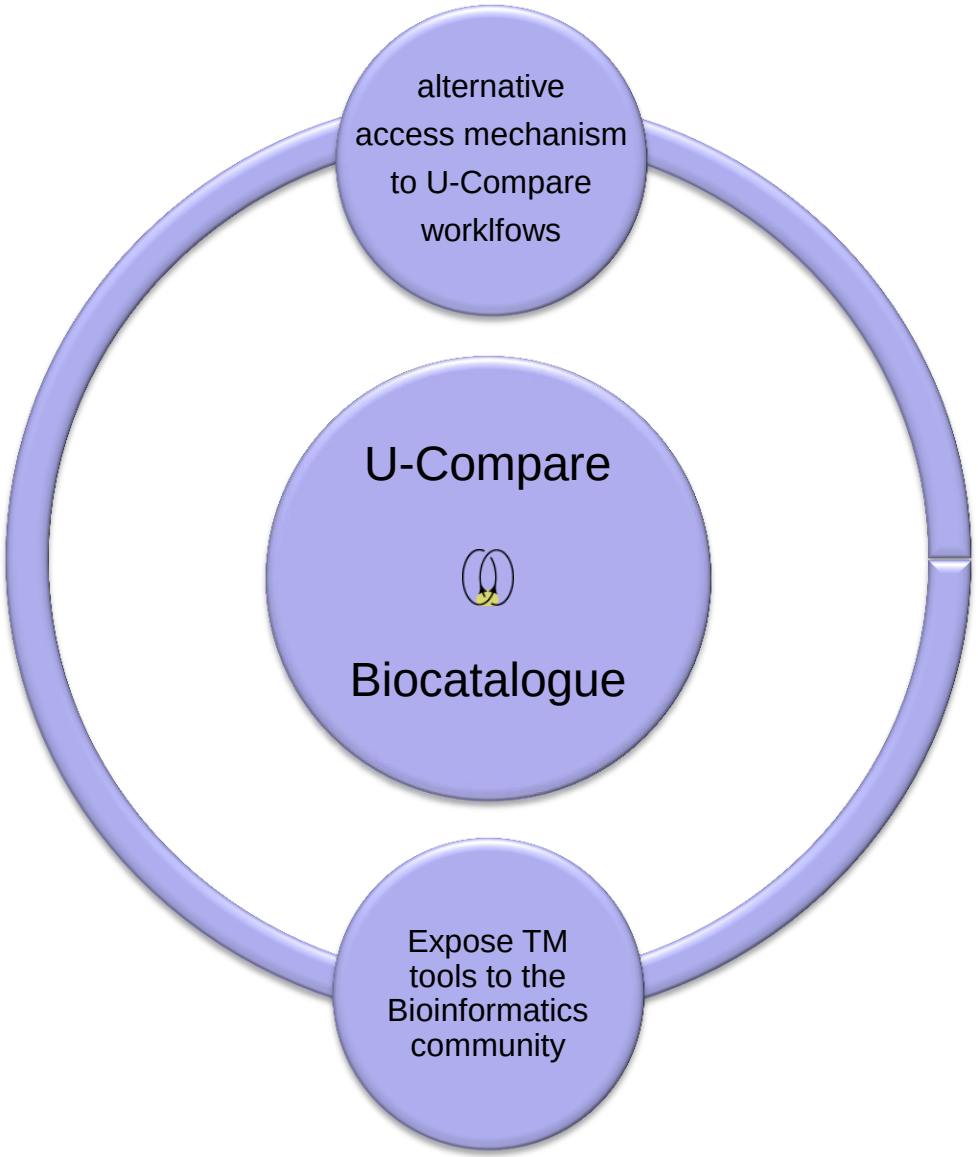
Biomedical NER

Web Service	Annotations
NeMINE	gene, protein
ABNER	protein,DNA RNA,cell line, cell type
GENIA	protein,DNA RNA,Cell line, cell type, token, chunk
DECA	protein,DNA RNA,Cell line, cell type, species
Yeast Metabliner	metabolites
OrgHub	organisms, habitats

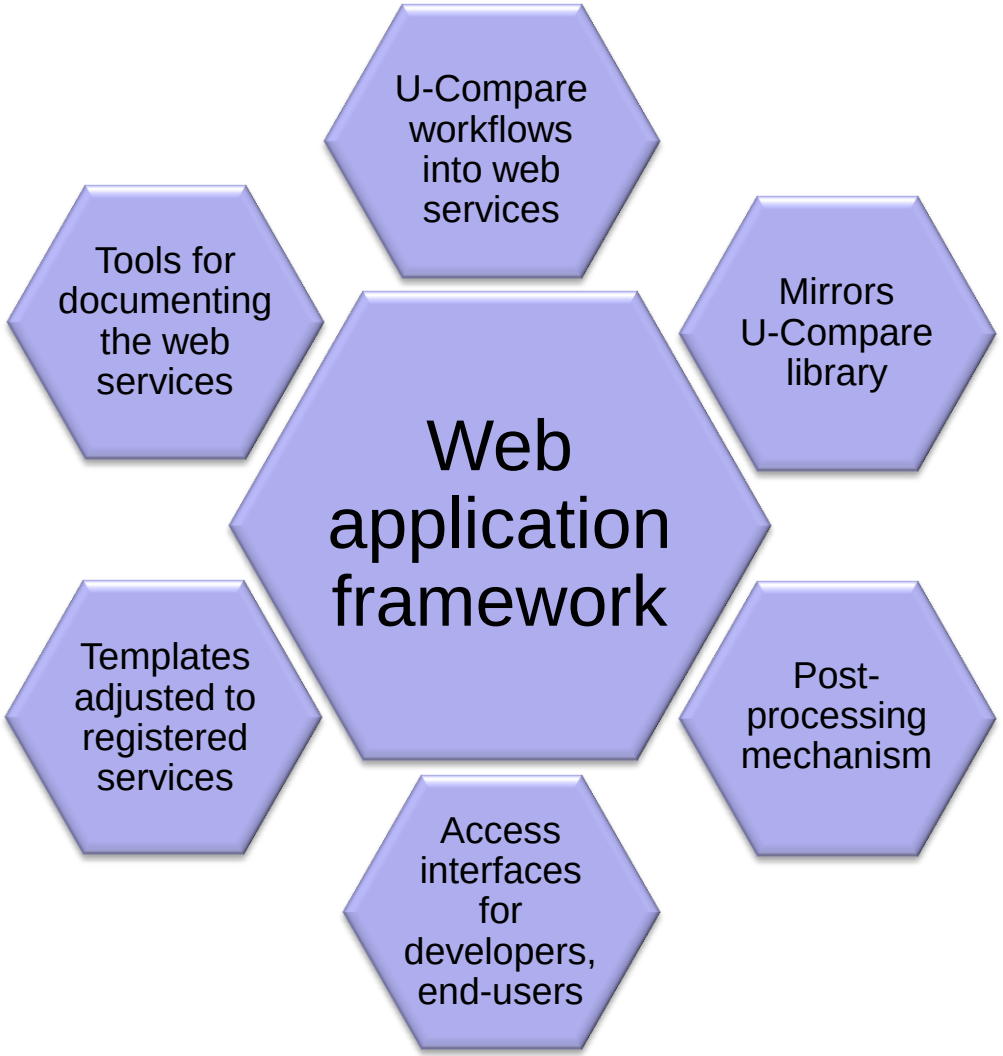
Chemical NER

Web Service	Annotations
OSCAR	gene, protein
OSCAR	chemical compound reaction, chemical expression, chemical adjective, ontology term

Conclusion & future work



Conclusion & future work



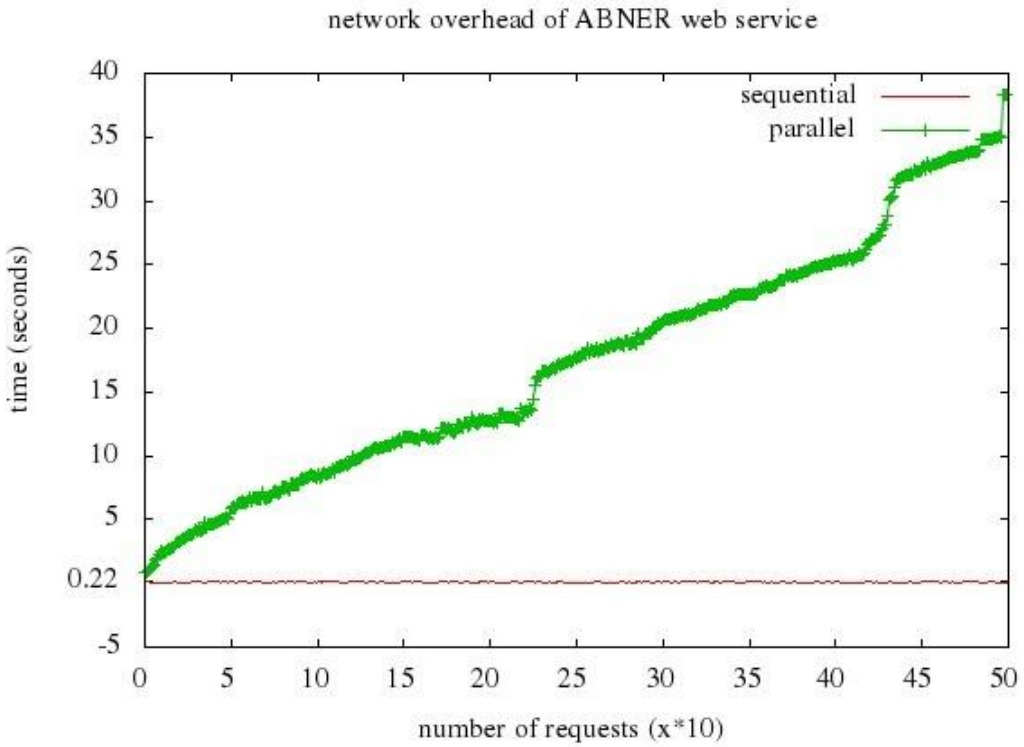
Conclusion & future work

Integration of the two frameworks is not automated yet

→ Embody framework into U-Compare

Framework presents delays in case of an overloaded network

XML is on of the ways that we use to model a web service




Questions